

Matchmaking Problems

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1 Introduction

In any scenario that pairings are made, there are a number of ways to optimize these pairings. We will cover a number of pairing problems and discuss different options to find optimal pairings in different scenarios. Before we can delve into these finer details, we will lay the groundwork in this introduction with definitions and necessary background.

1.1 Mechanism Design

All of the following pairing problems, we will rely on Mechanism Design, which is sometimes called "reverse game theory" in order to optimize the pairings. In each mechanism, we will consider n players and their preferences over A , a set of alternatives. We can also call these preferences orderings or utilities. The mechanism (or algorithm) which we will design in each case, will be used to create optimized pairings according to the individual problems at hand. In any case, the mechanism will take preferences of the players as input and output $a \in A$, the ideal alternative for a given player based on all preferences.

1.1.1 Defining Strategies

Definition 1. Strategy [3]

A **strategy** is a complete contingent plan for a player in the game.

Definition 2. Outcome

The **outcome** of a game is the ultimate result of a strategic interaction with one or more people, dependent on the choices made by all participants in a certain exchange. [2]

Definition 3. Dominating Strategy

A strategy B strictly **dominates** ($>$) strategy A : choosing B always gives a better outcome than choosing A , no matter what the other players do.

Definition 4. Weakly Dominating Strategy

A strategy B **weakly dominates** ($\geq$) strategy A : choosing B always gives at least as good an outcome as choosing A , no matter what the other players do, and there is at least one set of opponents' actions for which B gives a better outcome than A . [1]

1.2 Graph Conventions

In this paper, we will use graphs to demonstrate the preferences of the n players over A discussed above. Arrows between nodes will be used to represent different orderings, with arrows between orderings denoting which is more preferred. The more preferred ordering will be at the point of the arrow.

2 Auctions

The following actions are simple examples of mechanism design problems; it will be helpful to build up intuition and notation. The simplest auction we will consider is a single item, single round auction. In short, all players submit a single price without knowing the submitted prices of other players. The player with the highest submitted price will win the item. Players' bids are more optimized the closer they are to the bids below them while still winning the item and paying no more than what the item is worth to them.

2.1 Notation

- **Players** are given by i , their index
- $\hat{v}_i \in \mathbb{R}$, the **real value** of the auctioned item for player i
- $v_i \in \mathbb{R}$ is the **submitted value** of the auctioned item for the player i
- A game is **subsidy free** if the amount paid by a player i is given by p_i is nonnegative for all players

Definition 5. Strategy-proof

A game is **strategy-proof** if the submitted value equals real value is a weakly dominating strategy.

Notes 6. Lost Values

When we explore different strategies, we need to have a consistent metric by which we can explain what makes a strategy better or worse. We will refer to the "lost value" in given results of different auctions to compare them. In the case that a player wins the auction but pays more than their real value for the item, the lost value is the difference between the bidder's real value and submitted value. On the other hand, if the bidder loses, but the winner pays less than the bidder's real value then the bidder's lost value is the difference between the winner's payment and their real value.

Notes 7. A Note About Realistic Strategies

See that by our definition of lost value, it will never make sense for a player to bid more than their real value for a prize, since they would always prefer to lose out on the prize than take on lost value.

2.2 Examples of Auctions

We will explore two distinct auctions and walk through the different strategies the players/bidders can take to have optimal results.

Example 8. First Price Auction

A First Price Auction is the most common type of auction. All players submit their prices and the person with the highest submitted price wins the auction, paying their submitted price. A formalization is shown below with $f(v)$ representing the winner.

$$f(v) \in \arg \max_{i \in [n]} v_i \qquad v_i = \begin{cases} v_i & \text{if } i = f(v) \\ 0 & \text{otherwise} \end{cases}$$

Claim 9. This is an example of a game with no weakly dominating strategy.

Proof. Consider the case a player i 's bid is the highest i.e., they've won this auction, paying v_i , their submitted value. Consider then that all other players could have not bid/bid \$0. In this case, player i 's strategy will clearly change depending on their actions, as player i could lower their bid while still winning, which is clearly preferable, as they will have a lower lost value, according to Note 6. $\square$

Example 10. Second Price Auction (Vikrey Auction)

This auction is handled the same way as in Example 8, but the highest bidder pays the price of the second highest bid.

$$f(v) \in \arg \max_{i \in [n]} v_i \qquad v_i = \begin{cases} \max_{j \neq i} v_j & \text{if } i \in f(v) \\ 0 & \text{otherwise} \end{cases}$$

Theorem 11. The Vickrey Auction is strategy-proof.

Proof. We intend to show that the Second Price Auction is strategy-proof. i.e., we want to show that $v_i = \hat{v}_i$ is a weakly dominating strategy. Denote by t the highest bid made by players other than i . Begin by considering the case that $t < v_i$. In this case, the player i is content with their bid, as they win the prize while not overpaying. Now consider the case that $t \geq v_i$. If $t > \hat{v}_i$, player i would have no interest in bidding to win the auction, as they would take on lost value. If $t \leq \hat{v}_i$, player i should bid an amount in the interval $[t, \hat{v}_i]$. Thus, consider that player i 's bid should be in the intersection of all intervals $[t, \hat{v}_i]$, for all possible t . Thus, see that i should bid $\hat{v}_i$. Thus, $v_i = \hat{v}_i$ is a weakly dominating strategy. $\square$

3 Perfect Marriage Problem

In this example, we consider a set of monogamous, heterosexual men and a set of monogamous, heterosexual women. Each individual has preferences regarding each member of the other set (opposite gender). We intend to arrange married couples such that all married individuals are happy with their partner.

3.1 Definitions

Definition 12. Bipartite Graph

Let the bipartite graph G be defined by

$$G := (M, W, E)$$

Where M denotes the set of men, W denotes the set of women, and E denotes the set of edges.

Definition 13. Matching

A **matching** is pairing with elements from different groups.

Notes 14. Comparing Edges

All elements of M and W have individual orderings of elements of the opposite set. Put simply, Each woman has an ordering of their preferences in men and vice versa. If a woman prefers m_1 to m_2 , we will use the notation $m_1 >_w m_2$ to express this.

Definition 15. Blocking Edge

An edge mw is **blocking** a matching $\mu(w)$ if $m >_w \mu(w)$ and $w >_m \mu(m)$, i.e., if the woman w would prefer m to $\mu(w)$ and the man m would prefer w to $\mu(m)$. In the context of the problem, if an edge is blocking, the couple it represents should break up.

Definition 16. Stability

A matching μ is **stable** if $\nexists$ blocking edges.

Definition 17. Dominating (matchings)

A matching μ **dominates** a matching ν at W , if at $\mu(w) >_w \nu(w)$ for all $w \in W$.

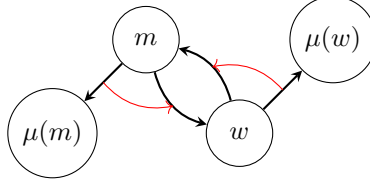


Figure 1: Graph representing the case that mw is a blocking edge

3.2 Gale-Shapely Algorithm

Definition 18. Gale-Shapely Algorithm

The **Gale-Shapely Algorithm** is one possible mechanism solution to the perfect marriage problem. The algorithm repeats the following steps in order until the pairings are unchanged by the algorithm:

1. Each woman proposes to the best available man (their top choice of the remaining options)
2. Each man rejects anyone but the best proposal (their top choice of the remaining options)

Theorem 19. Gale-Shapely Stability

Consider the matching μ given by the Gale-Shapely Algorithm. μ is stable.

Proof. Assume for contradiction that μ is not stable, which means that there is at least one blocking edge, and let mw be one blocking edge. In Figure 1, the preferences consistent with this blocking edge are shown by the bolded arrows and corresponding red arrows. If this is the case, then we can see that, according to the algorithm, w would propose to m rather than $\mu(w)$ first. As we see from the arrows showing m 's preferences, m have accepted a proposal from w , and by w 's preferences, we know that w would propose to m sooner than $\mu(w)$. Therefore, the only case in which m and $\mu(m)$ get married is the case that w moved on from m , which is not possible by her preferences. $\square$

The following two theorems give an intuition for how the Gale Shapely Algorithm is unique from other algorithms, but their proofs have been excluded from this DRP due to time constraints.

Theorem 20. Gale-Shapely Algorithm Dominates

Let μ be the output of the Gale-Shapely Algorithm, ν is an arbitrary stable matching. Then μ dominates ν at w .

Theorem 21. There is no strategy-proof mechanism for the Perfect Marriage Problem

4 Conclusion

Now that we have covered the Perfect Marriage Problem, we can expand our consideration to additional matchmaking problems, such as these College Admissions Problem in which we have a set of universities and a set of applicants. Each university has an individual quota of how many students they can admit and each student has their individual score on the standardized entrance exam. We hope to optimize each university's required minimum test score so that they accept as many students as possible while not exceeding their quota. In this problem, we are not only considering singular preferences of each individual, but also balancing accepting enough students to fill a college's incoming class while not sacrificing their selectivity. The College Admissions Problem serves as an example to show how the Perfect Marriage Problem can be the framework for many more matchmaking problems.

References

- [1] K. Leyton-Brown and Y. Shoham. *Essentials of game theory: A concise multidisciplinary introduction*. Morgan & Claypool Publishers, 2008.
- [2] M. J. Osborne et al. *An introduction to game theory*, volume 3. Springer, 2004.
- [3] J. Watson. *Strategy: An introduction to game theory. (No Title)*, 2002.